Chapter

The Inner Development Goals as an Innovative Approach to Sustainable Development: Conceptualization, Implementation, and Evaluation of an Experiential University Seminar for Holistic Sustainability Education

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Abstract

At the current rate, the UN Sustainable Development Goals (SDGs) will mostly not be met in time. An important reason is that we lack the inner capacity to deal with our increasingly complex environmental and societal challenges and to co-create a more desirable future. Recently, the Inner Development Goals (IDGs) initiative identified 23 crucial inner skills and qualities, structured along five dimensions. We contribute to the literature by developing, implementing, and evaluating an innovative experiential seminar for university students, which systematically covers all five IDG dimensions and touches on all 23 elements. It combines short conceptual inputs with a diversity of practical tools for experiencing and training the IDGs. The seminar targets an interdisciplinary mix of Master-level study programs, with students from many different nationalities. To tangibly illustrate connections between IDGs and SDGs and link to real-life sustainable behavior of students, the cocoa-chocolate value chain was used as a guiding example. Key themes covered were core values, character strengths, mindfulness, emotions, perspective taking, visioning, and changing habits. Among the key take-home messages that students indicated they would like to preserve, the most prominent ones are related to self-awareness and inner compass, followed by complexity awareness, perspective taking, appreciation, and optimism.

Keywords: inner development, sustainable development, experiential seminar, higher education, inner-outer transformation

1. Introduction

In 2015, the UN Member States adopted the 2030 Agenda for Sustainable Development. At the heart of the agenda lie 17 interlinked Sustainable Development Goals (SDGs) that aim to provide a shared vision of a prosperous future for people and the planet and guide strategic, collective action toward it. Yet, at the current rate, most of these goals will not be met in time. Even more worrisome, recent analyses show that between 2020 and 2023, progress in multiple areas has been halted or deteriorated, including in poverty eradication, ending hunger, and biodiversity conservation [1]. At the same time, the world is facing major socio-ecological crises. An important reason is that we lack the inner capacity to deal with our increasingly complex environment and to co-create a more desirable future [2]. Scharmer and Käufer [3] identify three key competencies for transformation: cognitive understanding of complex problem contexts, emotional intelligence, and the ability to convert intentions into action. Yet, conventional Education for Sustainable Development (ESD) largely focuses exclusively on the first, contributing to the well-documented intention-behavior gap [4]. In 2021, the Inner Development Goals (IDGs) initiative was founded, aiming to identify specific, science-based transformative skills and inner qualities needed to achieve the SDGs. In an elaborate interview process with renowned academics and practitioners, the initiative developed a framework consisting of 23 inner skills and qualities structured along five dimensions (Being, Thinking, Relating, Collaborating, and Acting) (**Figure 1**, [2]).

It has long been acknowledged that education is among "the most powerful and proven vehicles for sustainable development" [5 , p. 113]. In the context of inner transformation, education is a key to: (i) disseminating insights on the power of shifts in our individual and collective consciousness and their linkages to outer transformation to the broader public, and (ii) accelerating widespread training on specific skills and qualities. A central cornerstone of a new educational landscape that can support holistic sustainability education and the realization of the vision embodied in the

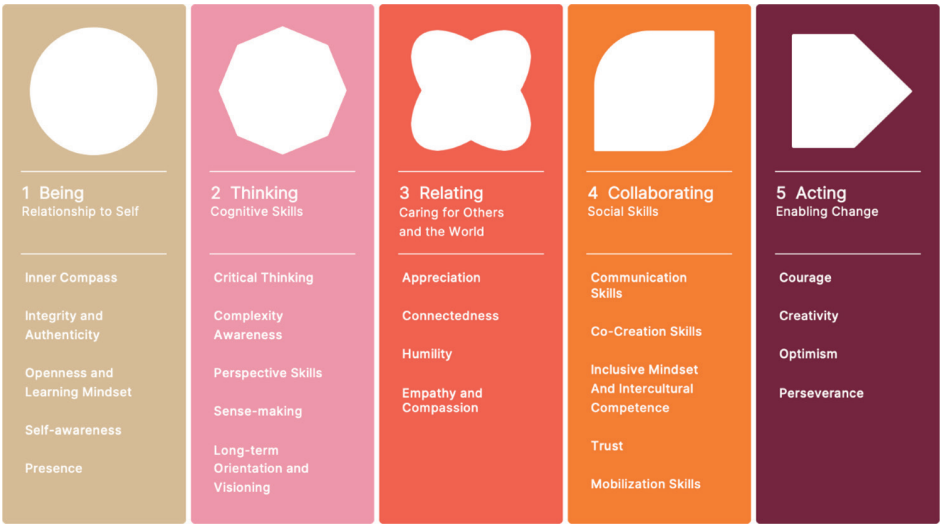

Figure 1.
The IDG framework. (Source: Available from: https://www.innerdevelopmentgoals.org/framework).

SDGs are innovative courses at universities or other higher education facilities. As they address and train young adults, those institutions play a major role in shaping future leaders and decision-makers, thereby having the potential to become a major motor for change. Unfolding this potential requires dedicated teaching staff and researchers who experiment with novel teaching approaches and design and implement innovative education programs that allow for transformational learning experiences along cognitive and socio-emotional dimensions.

Against this rising recognition of the central role and fundamental need for holistic approaches to ESD in higher education, in recent years a small number of pioneering programs have been developed. Often co-designed by international teams of researchers from education centers across the world, those programs experiment with various innovative pedagogical tools inspired by contemplative and reflective practices and relational ways of knowing, being, and doing [6–8]. Among the most prominent examples, Wamsler and colleagues developed a Master's-level course dedicated to linking inner and outer transformation utilizing a variety of mindfulness practices [5], building on prior theoretical and empirical research on possible linkages between mindfulness and sustainability and their current uptake in sustainability science, practice, and teaching (e.g., [6, 9–11]). Under the heading "Leading in Complexity", Ayers and colleagues [12] experimented with two tools under the reflective pedagogies umbrella (Portfolio and Pod) within a Master's level sustainability leadership course. They explored the benefits those tools hold for students and staff, as well as the challenges that were encountered. Developing and implementing an online course on EcoJustice, Walsh and colleagues [13] made great strides toward clarifying how a relational, justice-oriented approach could advance transformative education for sustainability.

Despite those recent advancements, the potential for and impact of training inner skills in ESD is still underexplored. To support the widespread uptake of innovative teaching programs, their evidence base demands further expansion. For example, a central insight emerging from previous efforts is that certain pedagogical tools and interventions often do not work for everyone to the same degree. While some students benefit greatly from a given practice, others find little help in the exercise or even struggle with it [12]. To enrich our understanding of what works best for whom and under which conditions, there is a need to experiment with and systematically explore the benefits and challenges of a broad variety of different tools and teaching techniques. Doing so will also contribute to greater fairness and inclusivity in classrooms. Additionally, while certainly emanating from the same underlying ideas of inner transformation as a deep lever of change and touching upon the broad dimensions of being, thinking, relating, collaborating, and acting, none of the aforementioned courses had been explicitly tailored to train all the 23 skills and qualities summarized in the IDG framework. Evaluation of those courses has mainly focused on how a certain innovative approach to teaching and learning (contemplative [5], reflective [12] and relational [13], respectively) may enrich ESD.

We contribute to the existing literature by developing, implementing, and evaluating an innovative experiential seminar for university students, allowing students to experience the IDGs from the inside. The seminar systematically covers all five IDG domains and touches on all 23 elements. To do so, it combines short conceptual inputs with a diversity of practical tools for experiencing and training the IDGs. The seminar targets an interdisciplinary mix of Master-level study programs, particularly Economics, Business Administration, and Environmental Systems Science, and was composed of students from a variety of nationalities spanning high-, medium-, and

low-income countries. Moreover, it centered around a guiding example (the cocoa-chocolate value chain) to tangibly illustrate connections between IDGs and SDGs and provide a connection to real-life sustainable behavior that students could easily relate to. Thus, in contrast to other existing case studies on innovative approaches to ESD, we followed a more outcome-oriented approach, starting from the vision to allow training on a broad spectrum of IDGs and then recursively exploring suitable pedagogical tools to train the respective skills and qualities in the expected participants of our seminar. In this way, the toolbox we deployed and tested in the context of ESD is more multifarious than those of similar pioneering courses. We draw from a more diverse body of literature, encompassing, among others, positive and social psychology, behavioral economics, contemplative practice, and participatory techniques from systems sciences.

The chapter is structured as follows. Section 2 presents the aims, framework conditions, and approach applied. Section 3 provides an evaluation of the results, based on an anonymous course evaluation survey, student workbooks, as well as lecturers‘observations. Section 4 concludes and identifies suitable adaptations.

2. Aims, framework conditions, and approach

This chapter presents a reflexive case study of an experiential seminar on “The inner development goals as an innovative approach for sustainability transformation”. The course was developed during the period from September 2022 to June 2023 and implemented in the summer semester (June–August) 2023. Below, we describe the aims, framework conditions, and methods applied. We structure our exposition according to the following three phases: seminar development, implementation, and evaluation.

2.1 Phase I: seminar development

The seminar was co-created by a core team consisting of an environmental and behavioral economist (Engel) and a systems scientist (Janssen), with selective logistic and thematic support from two further members of the group of Engel. The core team identified the following aims for the seminar. First, it should equip students with a general understanding of the IDG framework and its links to achieving the SDGs. Second, it should provide students with the possibility to experience the framework from the inside, i.e., by working on building and strengthening selected skills and qualities listed in the IDG framework themselves. Through this experiential approach, the seminar should go beyond the mere transfer of cognitive knowledge to also address aspects of emotional intelligence and the ability to convert intentions into action.

In fulfilling these aims, several framework conditions needed to be taken into account. The seminar formed part of the electives in three Master-level study programs at Osnabrück University: Economics, Business Administration, and the interdisciplinary Systems Science program “Environmental Systems and Resource Management”. Within the MSc program in Economics, which was recently converted to English, the seminar was part of a new Major in “Sustainability, Behavior and Environmental Policy”. To integrate well into these programs, it was important that the seminar established clear links to the social, ecological, and economic aspects of sustainable development. To maintain a solid grounding in science, it was decided

to select experiential tools and methods that are based on tested interventions from (positive or social) psychology, (behavioral, institutional, or environmental) economics, systems science, or neuroscience. Moreover, the combination of study programs implied a heterogeneous mix of students in terms of disciplines, nationalities, and prior knowledge. Successful completion of the seminar yielded five ECTS, out of an overall Master-level study program of 120 ECTS (with 90–100 ECTS in form of coursework). The general seminar module description was flexible in that it allowed for an applied, practice-oriented focus and could be conducted in regular weekly sessions or block format. The study program regulations set boundaries on the evaluation criteria to be used: The final grade was to be composed of 50% each of oral and written performance.

2.2 Phase II: seminar implementation

We implemented the seminar in the summer semester of 2023 and opened it to students from the three above-mentioned study programs. We set the maximum number of students to 26. Students had to apply for admission to the seminar by submitting their current performance record. It was announced that students from the Major "Sustainability, Behavior and Environmental Policy" and those close to finishing their studies would be given priority, but that also an interdisciplinary mix of students was to be a core element of this course. We opted for a combination of three short introductory sessions during the semester (90 minutes each), and three more intense day-long sessions during the semester break (from 10 am to 4 pm with a one-hour lunch break and several short breaks in between). The lecture team was composed of the core team (Engel, Janssen), with two short interventions by Dr. Marieke Baaken and logistical support from Dr. Fabian Thomas.

2.3 Phase III: seminar evaluation

Our evaluation of the seminar is based on three pillars. First, students had to submit electronic workbooks centered around predefined questions regarding key insights and reflections on seminar contents and exercises. These workbooks were the basis of the written performance grade. For the purpose of our evaluation of the seminar and to assure anonymity in the presentation of results, the answers to relevant questions were compiled and coded by the core team. Following a mainly deductive approach [14], the IDG framework functioned as a coding schema to unpack and systemize the workbook entries on students' key take-home messages and learn about the representation of different IDG dimensions in these. To this end, the workbooks were coded using the five overarching IDG dimensions as themes and their respective 23 subcategories. To limit subjectivity in the interpretation of the qualitative data and provide inter-coder reliability, the coding was done separately by both core team members; differences were then discussed jointly until a unanimous agreement could be found. To further enrich and refine the insights gleaned from the workbooks, in a subsequent step, inductive coding was used to identify additional patterns and themes emerging from the data [14]. Inductive coding also allowed us to make sense of students' (emotional) experience with the sustainability challenges along the cocoa-chocolate value chain and with the journaling task. For illustrative quotes from the workbooks, individual quotation-specific consent from the students for including these in an anonymous manner in this chapter was obtained.

We acknowledge the possibility that the workbooks provide a distorted picture since they were not anonymous and were a key basis of grading, which could have led to purposefully polished answers. Thus, to supplement our evaluation and cross-check the consistency of key results, we asked students to fill out an anonymous survey focusing on general aspects of the course and its structure, but also touching on some more in-depth topics similar to those addressed in the workbooks. The survey was conducted using Lime Survey and implemented toward the end of the last seminar session. Anonymity was assured by conducting the survey online and without using any student identifiers. Quantitative information from this survey was transformed into statistics, while open-ended answers to selected questions were coded before equally being transformed into statistics, sometimes complemented with selected quotes. Third, we complement these insights with our own observations as lecturers and co-learners.[1]

3. Results

3.1 Results phase I and II (seminar development and implementation)

3.1.1 Guiding example

To make the course contents more concrete and facilitate students' connection to the theme, we decided to use a tangible guiding example throughout the seminar. In general, many potential themes such as climate mitigation or environmentally-friendly consumption may be suitable for this purpose. We opted for the specific example of the cocoa-chocolate value chain as it provides for rich links to a broad set of SDGs (see **Figure 2**), and we could build on background from a research project in which both core team lecturers were involved.

3.1.2 Learning objectives

We defined the learning objectives as follows. At the end of the seminar, students should be able to: (i) describe the IDG framework, its background, motivation, scientific bases, and its links to achieving the SDGs; (ii) illustrate the relevance of SDGs and IDGs using the example of the cocoa-chocolate value chain; (iii) understand and critically reflect on their own role in collective outcomes; (iv) know and appreciate their unique self more (self-awareness); (v) understand others' perspectives more easily; and (vi) expand their strategy set for promoting the well-being of themselves and others in alignment with the SDGs.

3.1.3 Course structure and elements

Table 1 provides an overview of the course structure and elements, the main themes covered, and the key IDGs we aimed to address. With the exception of the

[1] As Walsh et al. [13, p. 1601] rightly remark, in experimental and transformative approaches to education "[t]eachers and facilitators should acknowledge that they are co-learners" who may not yet know which techniques work best or at all. Similarly, Wamsler [5, p. 125] highlights the importance of "the humility of not knowing". We agree with these authors and have communicated our seminar as a "joint learning journey" to our students.

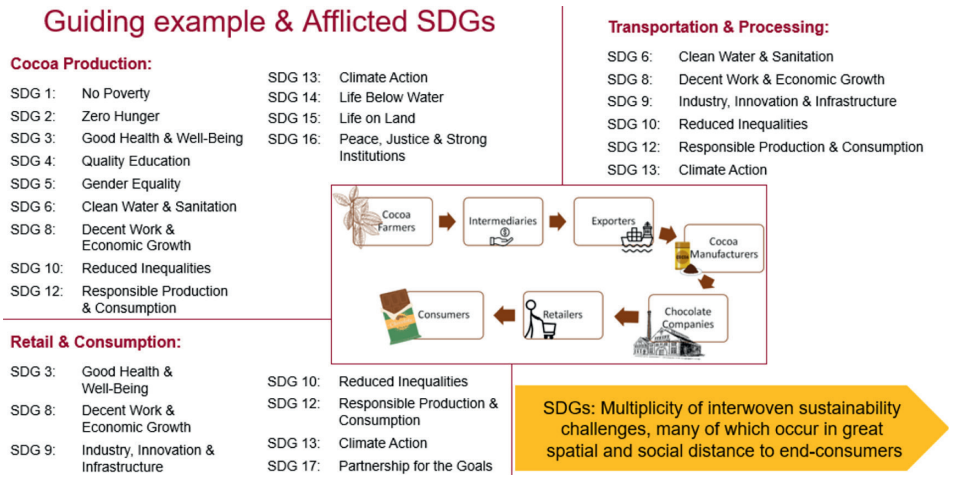

Figure 2.
Links between the guiding example of the cocoa-chocolate value chain and the SDGs.

Sessions	Main topics	Main IDG focus
Several	Overview of IDGs covered in the respective session	Sense-making
	Scientific evidence (S 2–6)	Complexity awareness, sense-making
	Sharing on exercises in subgroups and plenary (S 2–6)	Communication skills; trust; inclusive mindset & intercultural competence; connectedness
	Positive daily reflection (HW, S 1–5)	Self-awareness; appreciation
	Plant growing (HW)	Perseverance; appreciation; long-term orientation
	Mindfulness exercise (S 4–6)	Presence
1	SDGs, IDGs, & conceptual links	Sense-making
	Guiding example cocoa-chocolate	Complexity awareness
	Course setup and rules, incl. Rules for respectful communication	Communication skills
2	Core values	Self-awareness; integrity & authenticity
3	Character strengths	Self-awareness; integrity & authenticity
	Mission statement	Inner compass
4	Mindfulness	Presence
	Emotions	Self-awareness
	Perspective taking	Perspective skills; empathy & compassion
5	Deep listening	Communication skills; empathy & compassion
	Vision building	Long-term orientation & visioning; co-creation skills; creativity; optimism
6	Changing habits	Perseverance
	Overview of IDGs covered in the entire seminar	Sense-making
	Reflection on seminar & feedback	Communication skills

S = Session, HW = Homework.

Table 1.
Course structure and elements.

first (introductory) session, each subsequent session combined brief input from the lecturers with exercises and discussion. The first session provided introductory information by the lecturers on the SDGs, IDGs, and the guiding example of the cocoa-chocolate value chain, conceptual links between these, the general course setup, as well as rules of communication and interaction we asked everyone (including ourselves) to adhere to. Students were asked to choose one aspect of the cocoa-chocolate value chain and related SDG that motivated or moved them most. In the remaining sessions, the input from the lecturers detailed the IDGs addressed in the session, introduced the exercises and related concepts, and subsequently provided brief scientific evidence regarding the impacts of these IDGs and/or training exercises.

In terms of themes, sessions two and three focused on the exploration of individual core values and character strengths, their links to the chosen SDG, and the formulation of a personal mission statement.[2] The guiding example of the cocoa-chocolate value chain facilitated links to a variety of core values. These two sessions aimed particularly at the Being dimension of the IDG framework, which was seen as an important basis for the other dimensions [2]. This focus was also continued in session four with the introduction of mindfulness practice and a focus on emotion awareness/regulation.[3] The remainder of session four was dedicated to the practice of perspective skills, as a core element of the Thinking dimension, but also promoting empathy and compassion as key elements of the Relating dimension. This included an exercise in which students were asked to write a letter from the perspective of a cocoa-producing farmer from Ghana, based on a video showing the farmer‘s working conditions and resulting livelihood impacts.[4]

The focus on the Relating dimension was continued in session five with concepts and practices of deep listening, also training communication skills as a key element of the Collaborating dimension. The latter dimension (particularly communication, co-creation, and creativity) was further trained by making visioning a major theme of the seminar. In the related exercise, students were assigned to heterogenous groups of five to six participants and led through a process of participatory vision building, following the Seeds of Good Anthropocenes approach [26]. The visioning exercise was intended to also promote long-term orientation and visioning (Thinking) and optimism (Acting).[5] The visions were presented in the plenary at the start of session six. The subsequent main focus of session six was on the Acting dimension (perseverance), by providing tips for changing habits and an exercise in which students formulated a pledge on a small,

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[2] Core values were explored VIA a value card set and motivated by the role of values in motivating human behavior [15]. Character and signature strengths were explored based on the *VIA* test (Available from: https://www.viacharacter.org/survey) and motivated by their impact on various drivers of subjective well-being (see Available from: https://www.viacharacter.org for a list of supporting scientific evidence).

[3] Mindfulness practice took the form of short presencing meditations; emotions were explored in response to watching a video on socio-ecological implications of cocoa production conditions and using a list of emotion words adapted from [16, 17]. Both were motivated by their impacts on subjective well-being and their potential role in addressing the intention-behavior gap (e.g., [18–20]).

[4] The exercise was adapted from [21], and motivated by evidence that perspective-taking can increase empathic concern [22] and enhance pro-social and pro-environmental behavior (e.g., [23–25]).

[5] The visioning practice was motivated by evidence of impacts on how people see the world, what they expect from it, and what they deem possible [27], stakeholders‘motivation and commitment to work toward the jointly created vision [26], group identity and perspective taking (e.g., [26, 28]), and the co-creation of novel ideas [28].

concrete behavioral change.[6] As a common element across all sessions, the provision of scientific evidence supporting the themes and exercises was aimed at promoting complexity awareness and sense-making (Thinking).

Figure 3 provides more details on all exercises. The exercises were conducted individually and/or in groups of varying sizes and ultimately touched upon all 23 IDGs. In-class exercises combined self-reflection with exchange in small groups followed by plenum discussion (think-pair-share). The earlier group discussions were mostly in small groups of two or three to allow for some privacy and to slowly get comfortable with communicating personal insights with fellow students. Later exercises (starting with visioning) were done in larger (5–6 persons) groups that were composed by the lecturers to ensure heterogeneity in terms of study programs, gender, country of origin, English skills, and degree of prior participation in-class discussions. The exchange in subgroups and the plenary aimed to train various skills from the Collaborating dimension (communication skills, inclusive mindset, intercultural competence, trust), as well as connectedness (Relating).

Between sessions, individual-level homework assignments were given. Two of those stretched over the full seminar: (i) positive daily reflection, aimed to promote appreciation (Relating) and presence (Being)[7]; and (ii) growing a plant, aimed to train perseverance (Acting), appreciation for producers'efforts (Relating), and long-term orientation (Thinking). A recurrent element in the full-day sessions (sessions 4–6) were the short mindfulness exercises, thus continuing the training of presence. As the seminar was conducted for the first time, some short-run adjustments in timing were made; e.g., it was decided to extend the visioning exercise into the last session to allow for sufficient time to explore and discuss the co-created scenarios.

3.1.4 Diary and evaluation criteria

Students were asked to write down their reflections and insights from the exercises in a learning diary, a soft-cover notebook including key exercises and concepts. It was meant as a confidential course companion, in which students could note down their insights just for themselves. By contrast, the electronic workbook, which needed to be submitted individually by each student at the latest 2 weeks after the last seminar session and was the basis for 50% of the grade, contained questions on selected exercises only, assuring sufficient room for students' privacy.[8]

The grade for oral participation (50% of the total grade) was based on (i) physical attendance of all six seminar sessions (obligatory[9]); (ii) having done the homework exercises; (iii) active participation in the exercises and engagement with the contents, and (iv) abiding by rules of respectful communication. The grade for written performance (remaining 50% of the total grade) was based entirely on the electronic workbook.

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[6] Tipps were based on [29, 30]. Pledging was motivated by evidence of behavioral impacts [31, 32].

[7] The practice was adapted from [33] and motivated by the positive impacts of gratitude practice on subjective well-being [34].

[8] For example, while the exercises asked students to identify their top five core values and character strengths, the workbook only asked for revealing two.

[9] In the (one) case where a student could not attend one session, a written make-up exercise was defined and the student was required to pick up on the session contents in self-study, conduct the session exercises, and discuss these orally with the junior lecturer to simulate the in-class exchange.

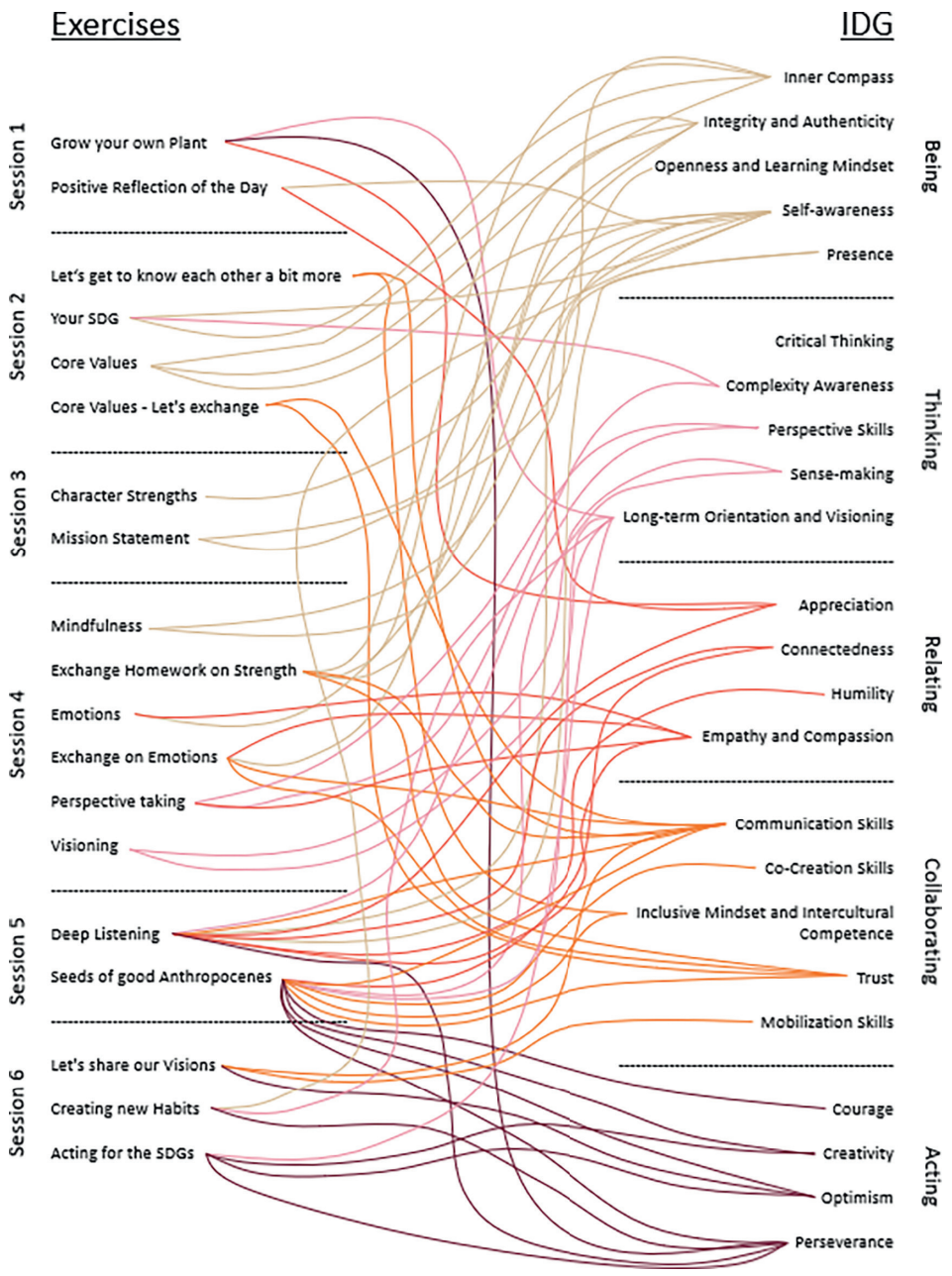

Figure 3.
Links between exercises of the seminar and individual IDGs.

3.1.5 Participants

Twenty-nine students applied for the seminar, twenty-six were accepted, while the other three were placed on a waiting list. We emphasized the importance of reading carefully a more detailed course description, which highlighted again that this seminar required a willingness to actively participate in a journey of self-growth and inner

development. Four students withdrew before the seminar started, which led to all students on the waiting list being offered a spot. One of these did not respond to the offer. Another three students did not show up for the seminar, yielding a final number of 21 actual participants (11 female and 10 male).

In terms of study programs, five participants were from the M.Sc. program in Business Administration, two from the interdisciplinary M.Sc. program in Environmental Systems and Resource Management, and the remaining 14 were from the M.Sc. program in Economics. Two students were in their first semester, 10 in their second, three in their third, three in their fourth, and three in their fifth semester. Six of the 21 participants were non-German natives, including students from Eastern and Southern Europe, South America, Sub-Saharan Africa, and two from Asia.

3.2 Results phase III (evaluation)

3.2.1 Students' overall course evaluation

Generally, the survey results indicated that the course was very well received. On a scale from one (very good) to five (insufficient), the best possible grade was given for the course by two-thirds of the seminar participants (mean = 1.38)[10] and for the lecturers by more than 90% (mean = 1.09). 76.2% of the students in the seminar would definitely, and the remainder probably, recommend the course to fellow students. 61.9% considered the course to be highly relevant for their future, another 33.3% somewhat relevant.[11] 90.5% of the participants indicated that they feel the concepts covered in the seminar can be applied to real-world situations or their (academic/work) context.

3.2.2 Students' background and perceived learning experience

As expected, students' backgrounds were heterogeneous. When asked about their level of prior knowledge on the subject of inner growth as an approach to sustainability transformations, 14.3% answered "very low," 38.1% "rather low," 28.6% "neither low nor high," 4.8% "rather high," and 14.3% "very high." This also affected the perceived learning experience, which 42.9% indicated as "very high," 23.8% "rather high," while 33.3% put "neither low nor high." (No one answered "very low" or "rather low.") Presence and appreciation are both goals of inner development. So what we mean here is that this heterogenity was evident particularly with respect to these two IDGs presence and appreciation. While some reported that they had already tried mindfulness meditation or gratitude practices before, for others it was an entirely new experience they needed some time to get familiar with. Particularly those who had never tried such exercises before highlighted the benefit they took from it. Several students also drew connections between what they had learned during the seminar and background knowledge they had already acquired through media or other courses. Some actively applied the theories and models they knew from their study program to the novel content (e.g., speaking of spillover effects,

[10] In evaluating the seminar, 66.7% of the participants gave the grade 1 (very good), 28.6% the grade 2 (good), and the remaining 4.8% the grade 3 (satisfactory). In evaluating the lecturers, 90.5% gave the grade 1 (very good), and the remainder 9.5% gave the grade 2 (good).

[11] Only the remainder 4.8% considered the seminar to be not relevant for their future.

or contemplating the potential of chocolate needing to become "something similar to a luxury goods market" to make people rethink their consumption patterns). Noteworthily, it was not only the disciplinary background but also personal life experiences that shaped students' perspectives on the content of the sessions. For example, one student mentioned that he grew up on a farm and therefore felt particularly moved by the social and environmental problems affecting the cocoa farmers. In line with this, he chose SDG 8 (decent work and economic growth) as the SDG that motivated him most, especially related to the cocoa-producing farmers. These familiar frameworks, be they inspired by personal and/or professional experience, became the lens through which the participants contemplated the IDGs and could make sense of the entangled complexity of the sustainability issues along the cocoa-chocolate value chain. Crucially, however, students did not remain within this initial, albeit helpful, but also static frame. Rather, through engaging in intercultural and interdisciplinary exchange and co-creation, students extended their horizons and learned to experience their own perspectives as merely one among many.

3.2.3 Value of intercultural and interdisciplinary exchange

The value of intercultural and interdisciplinary exchange emerged as one of the major perceived benefits of the course. In the survey, when asked what they particularly liked about this course, 38.1% of the participants emphasized the interaction and exchange with others, and 9.5% the diversity of the other participants and perspectives. For example, one student stated, "I really like the diversity that could be found among the participants in terms of the fields of study and so on since it allowed for wider vision of people's perspective and opinions." In the workbooks, several students noted that the interaction and exchange with the other participants greatly contributed to them enjoying and benefiting from the seminar. The openness and willingness of others to be vulnerable were, in turn, acknowledged and appreciated as prerequisites for this depth of interaction. This is again confirmed by the survey results. When asked what they particularly liked about the course, 38.1% emphasized the safe space created. The following statements from the survey illustrate this: "It really felt like a safe space, no judgement and openness." "I liked the Seminar all around. Especially the openness in the group. To speak about your thoughts and feelings with so many different people. That became very normal and comfortable." A few students emphasized the role of the lecturers and the block structure of the seminar in creating this safe space. By contrast, however, three students indicated in the survey that they felt somewhat uncomfortable in group discussions and activities. Looking at the open answers of those students, the underlying reasons appeared to be problems with the language, fear of speaking in front of the group, and feeling uncomfortable sharing quite personal things in front of a large audience. These concerns appear to relate particularly to the plenary discussions. This is illustrated by one of the same students indicating that (s)he particularly liked the exchange in small groups and suggesting to "maybe 'force' us to mix the groups up more 'cause in that way it might be easier to create trust 'cause you already worked with anyone." A certain initial reluctance or uneasiness when it comes to openly sharing emotions, feelings, and thoughts with others may also be linked to students' cultural backgrounds. In the oral feedback round at the end of the seminar, one student mentioned that in her home culture, it is rather uncommon for people to share what she felt were quite personal topics with others, even more so if those others are neither close friends nor family members.

3.2.4 Impacts on emotions

The intercultural and interdisciplinary exchange also had a notable impact on emotions. As part of the workbook, we asked students to report how they felt when thinking about "their SDG" at the beginning of the seminar vs. at the end of the course, and whether they could notice any changes in the emotional experience. While students selected a broad spectrum of different SDGs, ranging from social and/or ecological problems in the countries of production to sustainability issues on the consumer side, the majority noted exclusively negatively valenced emotions at the beginning of the seminar, such as sadness, frustration, bitterness, and helplessness. Emotion terms related to hopelessness were the most frequently mentioned (10 out of 21 students either explicitly used the term hopeless(ness) or noted that they were pessimistic, resigned, or felt discouraged). Those negative emotions largely remained present over the course of the seminar, and for the majority of students even deepened when gaining more in-depth insights into the harsh reality of cocoa production and the complex and interlinked nature of the social and ecological problems along the cocoa-chocolate value chain. However, over half of the students (11 out of 21) reported a notable qualitative change in their overall emotional experience over time. Specifically, the negative emotions were complemented by positive ones, such as gaining hope that things could change and a more positive future is possible after all. Seven students explicitly noted the exchange with the other students as the source for this hope. Five students referred to the power of engaging in a (collective) visioning exercise for helping to gain a more positive outlook on the future. Realizing that others, albeit coming from different cultural and disciplinary backgrounds, shared the same concerns and similar desire to work toward positive change induced a sense of connectedness that lent power and optimism.

3.2.5 Impacts of gratitude practice

Another exercise that exhibited a similar dynamic, though for different reasons, is the positive daily reflection. For the duration of the seminar (approx. 2 months), students were asked to engage in positive daily reflection in their learning diary on three questions that served as gratitude prompts: (i) What was positive today and how did I contribute to that?; (ii) What am I grateful for today?; and (iii) Who or what did support or inspire me today? Workbook entries indicate that the daily gratitude practice was by far no linear journey for students. While, in the end, most of them enjoyed it and recognized it as a beneficial practice, the majority struggled in the beginning, finding this kind of conscious reflection rather unfamiliar and challenging. Overcoming this initial hesitation, they soon became more familiar with the exercise, increasingly starting to notice small things in their daily lives to be grateful for. This phase of ease was, however, often imbued with moments of frustration of always noting down the same things. The repetitiveness was experienced as conflicting with the idea of appreciating the uniqueness of each day, the thinking being that in order for something to make a day special something would need "to be big and really positive". It was only a few weeks into the exercise that students eased their self-set aspirations, realizing that, in fact, it was often the small things and the persons in their life that gave each day its unique flavor, and that all these things were in fact not that small or ordinary after all. In the words of one student: "Writing down only positive reflections, regardless of having a 'good' or 'bad' day, made me realize how I took for granted some things that aren't as trivial as I thought. It also made me have an

eye on positive little things throughout my day that make me feel good. So this is also something I want to keep practicing."

Other perturbations to students' experience with the journaling task resulted from days in which a negative mood was prevailing and from stressful times during the semester or in their private lives. In these contexts, students found it hard to pay attention to things they appreciated during the day. Nine students noted that during those times they also simply lacked the time and/or mental resources to sit down at the end of each day, reflecting on and noting down what they were grateful for. During those times, journaling felt like "another task in an already task-loaded life", leading some students to temporarily skip the exercise. While the majority of those who reported having taken a break at one point took up the exercise again when stress ceased, one student did not come back to the journaling practice. For him, the initial reason for quitting the exercise had not been stress but rather a displeasure he found in the task itself. According to his workbook entry, he already had regularly engaged in consciously thinking about the things and persons he was grateful for prior to our seminar. Being asked to write those thoughts down on a daily basis felt like a chore, conflicting with the pleasant feelings of gratitude. This student closed by noting that he still finds much value in being grateful "[b]ut this just happens in my head and i think thats easier for me." Interestingly, other students reported the direct opposite experience. For them, it was exactly the part of bringing their thoughts to the paper that yielded a great benefit, partly to become aware and organize their thoughts in the moment, but also to capture them for the future. Borrowing the words of one of our participants: "So from now, whenever I felt down, I can uplift my mood by reading them all again to realize how blessed I am in this life."

When reflecting on their experiences with the journaling task, a small number of students (five out of 21) explicitly mentioned which of the three reflection prompts had been the most challenging for them to answer. Interestingly, this experience also varied greatly between students. Two noted naming other people or things that supported or inspired them as the most challenging, as "[i]f something inspires me it has to be really extraordinary and per definition that rarely happens" and "listing specific people by name felt very intimate and was a bit uncomfortable at first [...], since acknowledging their impact on my state of mind underlined my own fragility/vulnerability and dependence on these special persons." Three students struggled particularly with acknowledging their own contributions to positive things and events in their life, as their own actions felt "just like needed and not something [...] special to mention somehow." However, gradually gaining practice in reflecting on their own contributions, students started to feel more empowered and to develop a sense of agency. While not struggling per se with acknowledging their own contributions, several other students echoed difficulties with this prompt on those days when everything seemingly went wrong, nothing special happened, or they felt down or unproductive. Yet, with a bit of practice, they recognized that they had more power in determining whether they would experience a day as positive by consciously guiding their attentional focus. Specifically, they learned to not hinge their sense of self-worth and accomplishment exclusively on the achievement of something extraordinary, like major milestones and goals, but to also give themselves credit for "small" and "trivial" things.

Notably, the practice of writing down a positive reflection at the end of each day also changed how students experienced the day itself. Nine out of 21 students reported that they could notice themselves becoming more mindful over time and paying more attention to the small things that made them happy while those things

were actually happening. That is, gratitude became something they not only learned to practice ex post but started to consciously experience in the moment. Five students even noted that the practice changed how they were approaching their relationships, namely with conscious and voiced appreciation for the important people in their lives, thus indicating a shift in the cluster of relational skills.

3.2.6 Key take-home messages

In the workbooks, students were asked to describe two key take-home messages, i.e., main insights they would like to preserve. As an overall pattern, all of these messages transcended different IDG dimensions, the vast majority (85.7%) covering even three or more different dimensions. This indicates how inextricably the IDGs are intertwined.

Figure 4 indicates the frequency of specific IDGs being included in the students' take-home messages. Self-awareness and inner compass were the two most frequently referred to (by 80.6% and 47.6%, respectively). This is consistent with the emphasis we gave to the Being dimension in our course design. Self-awareness insights related to a variety of aspects, such as own values, strengths, and emotions, how to regulate emotions and how that impacts own behavior, prior judgments in communication, up to noticing changes over the course of the seminar with respect to confidence or general perspectives on life. Complexity awareness (Thinking), perspective taking (Thinking), appreciation (Relating), and optimism (Acting) were close thirds (each mentioned by 42.6%), indicating that the course successfully impacted all five dimensions. Worthy to note, each IDG was touched on by at least one take-home message, and within each dimension, it was one, or in the case of Thinking two, IDGs that received significantly more attention than the others (**Figures 4** and **5**).

As a recent study found that young people in Germany have the most pessimistic view of the future ever observed [35], we found the observed increase in optimism

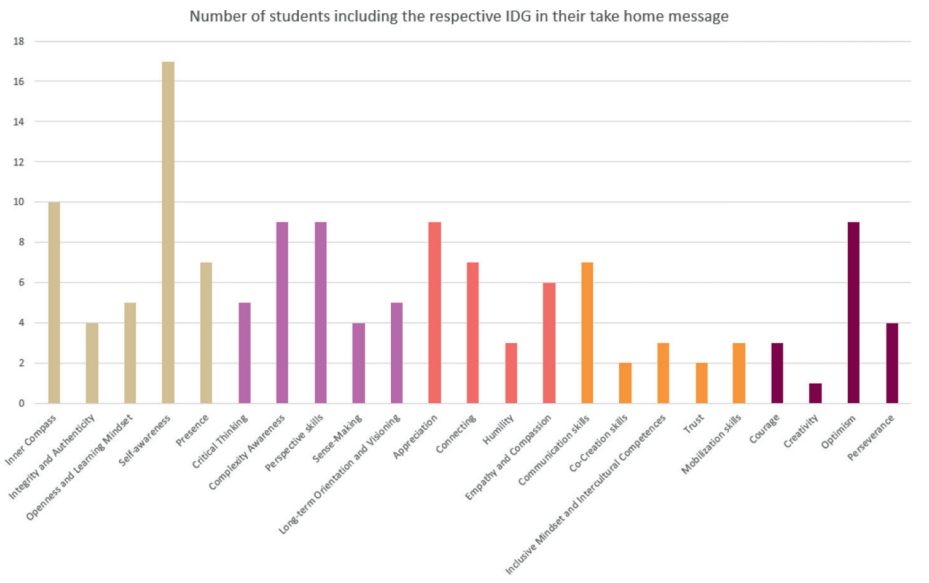

Figure 4.
IDGs included in students' key take-home messages.

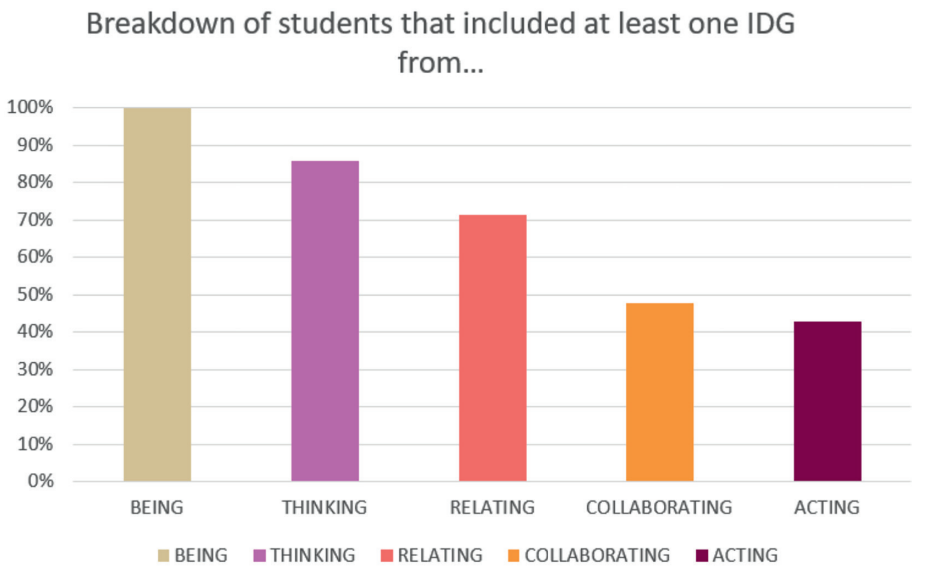

Figure 5.
IDG dimensions contained in students' key take-home messages.

particularly notable. To better understand the underlying processes, we delve more into the take-home messages of those nine students that indicated an (increase in) optimism. More than 40% referred to the power of positive visions, mostly for influencing collective outcomes, but in one case also for self-growth. For example, one student wrote: "If we only get sad and worried about things like climate change, it will make us feel hopeless and discouraged to introduce changes in our lifes (because we think it would not matter anyway), making situation worse. If we are optimistic, we feel motivated to change the world for the better". Another stated: "you can find solutions and compromises in every constalation of people and situation. In that seminar we were so many different people from different countries, backgrounds, ideas and feelings. But nevertheless we managed to develop solutions." The same percentage (44.4%) highlighted the realization that they themselves could make a difference, by altering their own behavior and thereby inspiring others. For example, one wrote: "I learned that living in peace with yourself, improving yourself, your awareness, your believe, your character and so on, can have a big impact on other people. [...] Because others see how you life and how good your feelings are. They maybe get inspired or influenced by this and adapt or adjust their behavior. This [...] gave me a lot of hope and joy when looking in the future." A third of the students whose take-home messages related to optimism referred to the feeling of not being alone with their worries and/or willingness to change, while another highlighted gratitude as a lever for adopting a less pessimistic view about the future. Thus, we find that the increase in optimism appears to result from the power of positive visions, an increased feeling of self- and outcome efficacy, an increased feeling of connectedness, and knowing tools to regulate negative emotions. While a link to optimism was explicit in the key messages of 42.9% of all participants, those underlying mechanisms, as well as the above-mentioned reflections on emotions, were observed also in many of the others' take-home messages. A further potential impact related to optimism, for which we have only anecdotal evidence, is that engaging with the concept of the IDGs may have

an impact on perceptions regarding the nature of the human being. In oral conversation with one participant, approximately 9 months after the seminar ended, she told us that the seminar changed her view on the human being, from being the selfish one usually portrayed in economics, to a more complex being that is inherently capable and willing to cooperate. Similarly, one student wrote in his take-home messages: "What I am going to take away from this seminar is that even in a tough business world, sustainability and personal development play a big role. It is not all about making money or having money; it is about the immaterial things in life, like emotions and relationships."

3.2.7 Spillover effects beyond the seminar duration

The workbook entries provide indications for the potential longer-term impacts of our seminar. In their answers on the positive daily reflection, nine out of 21 students stated that they believe that the effects of the gratitude practice will partly carry over after the official end of the seminar, having changed their subjective experience and attitude more permanently. Furthermore, tapping more into the insights consolidated in the take-home messages, we find that several students noted that they consider the practices learned during the seminar, particularly gratitude practice and mindfulness meditation (each mentioned by seven out of 21 students), as valuable tools to better cope with stressful times and negative emotions in the future. Moreover, all except for three students included in their take-home messages some kind of a pledge for the future, although this was in no way required. Examples of such pledges include promising oneself to show up more authentically in everyday life, to include short mindfulness practices in their routines, or to act more consciously when buying chocolate. This embodies a leap from acquiring new insights to making them meaningful in the context of one's own story by translating them into action plans for one's life.

3.2.8 Insights regarding course design and tool selection

To prevent the description of our results from becoming too lengthy, we will only briefly touch upon some further, more general insights regarding the course design and selection of teaching tools we gleaned from the survey as well as our own observations and experiences during the seminar. Survey responses indicated that participants particularly liked the experiential format; e.g., one wrote "I love the way the ideas being transmitted. Somehow talking about the sustainability or inner growth maybe very theoretical and hardly to get. But by the way of taking relevant actions for example growing salad tree or writing daily positive reflection helps me gradually realize the true concept behind it." 19% particularly liked the (opportunity for) self-reflection, 14.3% the topic of the IDGs (providing a new perspective), and 9.5% emphasized the diverse formats used in the course (learning diary, small group exercises, plenum discussion, etc.). When asked which specific exercise they liked the most and/or has remained in positive memory (multiple answers allowed), the majority (57.1%) named visioning, followed by the VIA strengths test (28.6%), the exploration of core values, and the work in small groups (19% each). Furthermore, the positive daily reflection, the salad planting, mindfulness, and the positive portfolio were each listed by 9.5% of the participants. More than 95% indicated that the (interactive) activities and discussions kept them engaged throughout the seminar and that the group activities were well-structured and conducive to meaningful

collaboration.[12] When asked for suggestions for improvement, three students (14.3%) each suggested increasing the ECTS of the course to allow for more depth and to create diverse subgroups already earlier. Other relevant suggestions by individual students included adding a timeline for the full-day sessions, explaining the IDG relevance for solving real-world problems a bit more, and presenting some more details on solution approaches.

4. Discussion and conclusions

Our seminar combines short conceptual inputs with a diversity of practical tools for experiencing and training the IDGs. Our approach differed from other existing innovative approaches to ESD [5, 12, 13] in that we followed an outcome-oriented approach, starting from the vision to allow training on a broad spectrum of IDGs, and then recursively exploring suitable pedagogical tools to train the respective skills and qualities in the expected participants of our seminar. Moreover, in order to tangibly illustrate connections between IDGs and SDGs and to provide a connection to real-life sustainable behavior that students could easily relate to, our seminar centered around a guiding example (the cocoa-chocolate value chain). As a consequence, the toolbox we deployed and tested in the context of ESD is based on a more diverse body of literature than those of similar pioneering courses. To our knowledge, this was also the first experiential seminar on IDGs explicitly targeted to students from economics and business administration.

Our aim was to equip students with not just a cognitive understanding of the IDG framework and its links to achieving the SDGs, but also provide them with the possibility to work on building and strengthening selected IDGs themselves. Our results indicate that students indeed felt such an impact. Their statements on the two key take-home messages indicate that particularly self-awareness, inner compass, complexity awareness, perspective taking, appreciation, and optimism were impacted, also showing that the course successfully addressed all five IDG dimensions. We thereby contribute to expanding the evidence base of innovative and integral teaching approaches, e.g., called for by Wamsler [5], as well as advancing understanding of how specific practices from the reflective toolbox could contribute to holistic sustainability education, e.g., called for by Ayers et al. [12].

One major finding, in line with the results of Ayers et al. [12] and Wamsler [5], is the value of having a diversity of student backgrounds, in our case in terms of disciplines and nationalities. Our seminar was targeted at an interdisciplinary mix of Master-level study programs in Economics, Business Administration, and Environmental Systems Science. Moreover, it was composed of students from a variety of nationalities spanning high-, medium-, and low-income countries, and four continents. Being confronted in group exercises and plenary discussions with this diversity in backgrounds and resulting perspectives sparks new ideas in students and inspires them to reflect on their own perceptions and experience. As such, diversity became a key motor for enhancing perspective skills, trust, and connectedness. In

[12] When asked whether the (interactive) activities and discussions kept them engaged throughout the seminar, 95.2% answered Yes, the remainder 4.8% answered No. The same distribution of answers was also given to the question of whether the group activities were well-structured and conducive to meaningful collaboration.

addition, we could observe an impact on optimism resulting from the observation that, despite the different backgrounds, it was possible to develop joint visions of a desired future, a finding that is in line with general findings on collective vision building (e.g., [26, 28]). Further reasons for an increase in optimism were the increase in connectedness, as well as an increased feeling of self- and outcome efficacy, knowledge of new tools to regulate negative emotions, and possibly also altered perceptions regarding the nature of the human being. We consider these insights to be an important and encouraging result in light of the generally observed increase in pessimism among young people [35, 36]. Moreover, as existing approaches to ESD indicate, positive emotions could also contribute to lowering overall stress levels experienced by students [5]. In terms of course design, the students' reflections on how their emotions toward the guiding example developed over the course of the seminar highlight how important it is to counterbalance moments dedicated to more in-depth insights on the lived realities of socio-environmental challenges or the complex web of intertwined causes with exercises that support a more positive outlook onto presence and future (e.g., gratitude practice, positive portfolio, and visioning).

There were a number of other factors that supported the achievement of our objectives. We benefited from flexibility in module descriptions and the embedding of the module in three study programs, as well as the fact that one of these programs was an English-speaking program, attracting students from many different nationalities. Where such conditions do not hold, additional steps may be needed, e.g., co-creating joint seminars offered in more than one study program.

In dealing with inner growth and emotions, a topic very uncommon in our usual study programs, the creation of trust and a safe space highlighted in our students' feedback was likely paramount for the success of the course (see Refs. [12, 13]). Upon reflection, we hypothesize that the following factors promoted the creation of that space. First, we formulated and shared rules for respectful communication right from the start, and made abidance by them a part of the oral grade. Second, while it is common in the German education system to address students by last name, we proposed from the start to use the informal "you" and first names in both directions throughout the seminar, arguing that this was a seminar in which we all encounter each other as human beings jointly engaging in a journey of inner development. Third, we as lecturers consciously aimed to show ourselves as authentic human beings, sharing also insights on our own inner development (e.g., on our values, strengths, mission, and difficulties with being mindful). Fourth, a few students opened up fast, sharing personal thoughts and emotions in reaction to the early exercises, which likely encouraged others to open up as well. This was likely fostered by us lecturers responding in a very appreciative, supportive, and related manner to the students' personal insights. These hypotheses are in line with insights from previous ESD courses highlighting the importance of the quality of relations among students and between students and teaching staff for the emerging learning experience and outcomes [12, 13]. For our seminar, evidence for our hypotheses is many student comments noting that they enjoyed and felt empowered and inspired by the kindness, openness, and respect they received from the lecturers. Despite these efforts and the generally very positive feedback on trust and safe space, there was a small number of students for which the plenary sessions did not convey these qualities. We thus find important the suggestion of one of these students to, in the future, gradually increase the size of the groups for the exercises, in order to allow a build-up of trust and confidence for everyone.

In line with findings on other ESD courses (e.g., [5, 12]), specific exercises had differential impacts on the heterogeneous set of students. This implies a need for a

mixed, multifaceted toolbox as well as flexibility on behalf of the teaching staff. For example, we modified the perspective taking exercise in the course of the seminar session in response to some students finding it challenging to take the perspective of a Ghanaian farmer; the modified exercise allowed stating own assumptions first and/or do the exercise from the perspective of the farmer's son who was a university student.

Some further insights relate to the IDG framework. We found that all exercises as well as the students' take-home messages addressed more than one IDG, often from different dimensions. The intertwined nature of the IDGs has been acknowledged by the IDG initiative.[13]

In addition, when trying to link students' take-home messages to specific IDGs, we faced a challenge that is often criticized as one of the IDGs framework's greatest weaknesses: The framework is relatively simple, the individual skills and qualities only broadly defined, and often partly overlapping. Answering this critique, the initiative notes on its website: "The framework's simplicity is a design principle, both keeping it easy to communicate and relatively 'naked' or decontextualised, ready to be re-contextualised within specific training or personal development approaches [...] This adaptability allows it to be tailored to diverse cultural and societal contexts, ensuring its relevance across different settings." [37]. We experienced our discussions on contextualizing the framework for our purpose as inspiring and valuable, benefitting from each other's perspectives and exchange around them. Very much in the spirit of the IDGs, engaging in mutual perspective taking and collective sense-making of how the skills and qualities could be contextualized in our setting helped us to engage with our ingrained beliefs on what inner development means at different levels, become aware of our own blind spots, and support each other in overcoming them. We thus conclude that the framework's "relative nakedness", while imposing challenges in the application, has strong merits.

We close with thoughts on the limitations of our study and further research directions. A limitation of our evaluation is that the student workbooks could potentially be biased. However, we found that the anonymous survey yielded a picture that is very consistent with the feedback contained in the workbooks. This makes us confident that the workbook and survey together provide a sound foundation for our insights. Nevertheless, to provide a more rigorous testing of our insights, a more systematic impact evaluation would be helpful. The most realistic implementation option in our view would be a survey-based before-after comparison, based on existing scales from social psychology. Where possible, this could be complemented by conducting the same survey also in a control group of students who applied to the seminar, but did not get a spot. To further test our finding on potential spillover effects beyond the seminar duration, an ideal evaluation would also systematically go beyond the duration of the seminar to assess the persistence of effects in the medium-term.

By nature, and in line with other studies of this type, our study could only document the impacts of our seminar for the given setting, and thus the results cannot necessarily be generalized. We are convinced that the findings will not only help ourselves to revise the seminar, but can also provide guidance to others interested in designing and implementing a seminar dedicated to inner development. It would be interesting to delve deeper into the differential impacts of the seminar by study programs and student background knowledge. One aspect that would be particularly

[13] For example, in the deep dive into the framework available at the initiative's website.

interesting to assess with respect to economics and business administration programs would be to systematically assess - through in-depth interviews or as part of the proposed survey-based evaluation - the generalizability of the anecdotal observation that the course changed students' perspective on human nature.

Closing with a personal note, we find that conducting this seminar has been by far the most rewarding and remarkable experience in our teaching careers.[14]

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Conflict of interest

The authors declare no conflict of interest.

[14] which for the senior lecturer spans 27 years.

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